MILITARY SPECIFICATION

INTERCOMMUNICATION STATION, TYPE LS-540/SIC,

NAVAL SHIPBOARD

1. SCOPE

1.1 Scope.- This specification covers a type LS-540/SIC, intercommunication station for use aboard Naval ships.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on data of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-27 - Transformers and Inductors, Audio, Power, and High Power Pulse, General Specification For.
MIL-C-62 - Capacitors, Fixed, Electrolytic DC, Aluminum, Dry Electrolyte, Polarized, General Specification For.
MIL-C-915 - Cable, Cord and Wire, Electrical, Shipboard Use.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements For.
MIL-C-3965 - Capacitors, Fixed, Nonsolid Electrolytic, Tantalum, Foil and Sintered-Slug, General Specification For.
MIL-R-5757 - Relays, Electrical, Excluding Thermal, For Electronic and Communication-Type Equipment, General Specification For.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-E-17362 - Electronic Repair Parts Requirements, Procedures for Provisioning Technical Documentation and Stock Number.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-C-26655 - Capacitors, Fixed, Solid Electrolyte, Tantalum, General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

- ENGINEERING EXPERIMENT STATION REPORT

060025 - Proposed Specifications for the Illumination of Interior Communication Instruments and Other Visual Display Systems.

(Application for copies should be addressed to the U.S. Navy Marine Engineering Laboratory, Annapolis, Md. 21402.)

FSC 5835

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OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 Preproduction sample.- A preproduction sample shall be furnished for the tests specified in 4.2 (see 6.3).

3.2 Description.- The intercommunication system shall provide two-way voice communication between the intercommunication stations included in the system on either one of two intercommunication channels provided. In addition, the equipment stations shall provide the operator with the capability for two-way communication over any one of three radios installed aboard a ship. Each station shall be designed for use with a headset—microphone as specified herein.

3.3 General features.- The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification the requirement of this specification shall govern.)

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Aluminum
Magnesium
Iron and steel
Nonferrous material (except aluminum)
Zinc
Springs (material)
Other metals
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds

Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Locking devices
Washers
Parts, electrical—general
Requirements for semi-conductor devices
Mounting of semi-conductor power devices
Heat sink
Electron tube or capacitor sockets
Variable resistors
Transformers and inductors
Relays
Electrical tapes
Batteries

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Enclosure—degree of (see 3.5)
Enclosure—mounting
Cable entrance plates
Threaded devices
Rounded corners and edges
Drilled and tapped holes
Welding
Temperature and humidity
Shock, vibration and inclination (vital vibration test required)

Shielding and radio frequency noise reduction
Soldering
Electrical parts mounting
Terminal boards, connectors and terminals
Wiring
Color coding
Electrical insulation
Dielectric strength and insulation resistance clearances
Drawings—general
Drawings—preliminary
Drawings—working
Interchangeability and standardization
Manuals

Repair parts (electronic)
Designation and marking
Item names and nomenclature
Workmanship and general examination

3.4 Size and weight.- The weight of the equipment station shall not exceed 5 pounds. The equipment size shall not exceed the following maximum dimensions (excluding control knobs and handles):

Height - 5 inches
Width - 6 inches
Depth - 7 inches

3.5 Enclosure.- The equipment enclosure shall be designed for mounting in a panel or on a bulkhead. The enclosure shall be made from aluminum material and shall be dripproof in accordance with MIL-I-983. The equipment shall be designed to allow removal of the front panel including the amplifiers and control circuits from the case of the unit for servicing. The front section shall be secured in the equipment station enclosure with knurled thumb screws with slotted heads. Mounting holes shall be provided in the back of the enclosure for bolting the unit to a bulkhead. The location of the holes shall be such that a socket wrench can be used to install the mounting bolts. The equipment technical manual shall include instructions for mounting the equipment station in a panel or console.

3.5.1 Gaskets.- Gaskets shall be used, if necessary, for making dripproof seals between the case and front panel assembly.

3.5.2 Drainage.- Drainage shall be provided for any panel recesses which otherwise would be susceptible to collection of water.

3.5.3 Alignment guides.- Alignment guides shall be provided to assure proper alignment between mating plug and jack connectors. The guides shall support the unit during insertion and removal intervals and while in the case. The guides shall prevent any damage to the plug-in connectors or any part of the unit during insertion and removal of the unit from its case.

3.6 Front panel.- The front panel configuration shall be in accordance with figure 1. Handle grips shall be provided on the front panel to facilitate withdrawal of the unit from its case.

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3.7 Servicing.- With the unit removed from its case, all components shall be readily accessible for ease of replacement. When the unit is removed from its case for servicing, all component parts shall be protected from possible damage when the unit is placed on any of its sides.

3.8 Electrical connections.-

3.8.1 Ship's wiring connections.- A terminal board shall be provided for connecting ship's wiring to the equipment. Sufficient terminals shall be provided on the terminal board for all necessary wiring connections.

3.8.2 Plug-in connectors.- Multi-contact plug and jack connectors shall be provided which will disconnect the unit from ship's wiring when it is removed from its case. The plug shall be mounted on the unit and the jack shall be mounted in the rear of the case such that the plug engages in the jack when the unit is placed in its case. The connectors shall employ a floating type of mount to assure positive alignment and positive connection.

3.9 Headset—microphone jack.- A type U-79/U headset-microphone jack, Amphenol type No. 165-7J or equal, shall be provided on the front panel. Wiring to the jack shall be such as to permit use of headset-microphones wired as shown on figures 2 and 3.

3.10 Selector switches.- The selector switches shall be lever key type switches with spring pile-ups. The lever shall actuate one set of contact springs when operated in one direction, actuate another set when operated in the other direction and release both sets when in the normal or center position. Switches shall be of the non-locking type and shall be treated for protection against high humidity. The equipment shall be capable of operating as specified in 3.16 after being subjected to the selector switch life test specified in 4.6.5.

3.11 Indicator lamps.- An indicator lamp shall be provided above each radio selector switch. The indicator lamp shall be illuminated when the associated radio is switched to transmit condition by any equipment station in the system having capability for transmission on the radio. The indicator lamps shall be subminiature, omnidirectional, red plastic dome types which use a T-1-3/4 lamp. The indicator lamps shall be energized by the 12 volt d.c. supply contained in the radio transmitter.

3.12 Illumination.- Front panel illumination shall be in general accordance with the requirements of Engineering Experiment Station Report 060025 Proposed Specifications for the Illumination of Interior Communication Instruments and Other Visual Display Systems. All selector switch function and position designations shall be illuminated. Illumination lamps shall be located to permit replacement of lamps without removing the unit from its case. The level of illumination shall be controlled by a continuous type control which is accessible with the unit removed from its case and provides for variation of the illumination level between the normal brillance and minimum brilliance levels specified in 3.12.1 and 3.12.2.

3.12.1 Normal brilliance.- Normal brilliance of switch designations shall be that level which is easily visible at a distance of 18 inches from the unit under average office lighting conditions.

3.12.2 Minimum brilliance.- Minimum brilliance shall be that level at which the designations are easily readable at a distance of 18 inches in a completely darkened room. This level shall be determined after a dark adaptation period of 10 minutes.

3.13 Transformers and inductors.- Transformers and inductors shall be grade 5, Class S, life expectancy X, as specified in MIL-T-27.

3.14 Relays.- Relays shall conform to MIL-R-5757 and shall be of the hermetically sealed type. Relay contacts shall be of the bifurcated type.

3.15 Capacitors.- Tantalum electrolytic capacitors conforming to MIL-C-3965 or MIL-C-26655 shall be used in the amplifier assemblies. Electrolytic capacitors in accordance with MIL-C-62 may be used in power supply filter circuits only.

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3.16 Operational features.-

3.16.1 Operation for communication on radio.-

3.16.1.1 Transmit and receive operation.- Operation of the equipment station for transmitting and receiving shall be the same for each of the three radio transceivers the station is capable of selecting. Operation of the station when transmitting and receiving over RADIO 1 shall be as specified in 3.16.1.1.1 through 3.16.1.1.6. Operation for transmitting and receiving over RADIO 2 or RADIO 3 shall be as specified in 3.16.1.1.1 through 3.16.1.1.6 except the RADIO 2 or RADIO 3 selector switches shall be operated to the TALK position in lieu of the RADIO 1 selector switch.

3.16.1.1.1 RADIO 1 selector switch shall be set to the TALK position.

3.16.1.1.2 RADIO 2 selector switch, RADIO 3 selector switch, and the INTERCOM selector switch shall all be set to the OFF position.

3.16.1.1.3 Incoming audio signal from RADIO 1 receiver shall be amplified and applied to the station headset.

3.16.1.1.4 With the microphone push-to-talk switch operated, the audio signal from the station microphone shall be amplified and applied to the audio input of RADIO 1 transmitter.

3.16.1.1.5 Operation of the microphone press-to-talk switch shall cause the RADIO 1 control circuit to be closed which in turn switches the radio to the TRANSMIT mode and illuminates the RADIO 1 indicator lamp at all stations capable of operating on RADIO 1.

3.16.1.1.6 Operation of any radio selector switch to TALK position shall open the transmission path from the microphone amplifier output to the INTERCOM selector switch. This prevents coupling of the intercom transmission line to the radio transmitter audio input, a condition which could occur if both the intercom line and the transmitter audio input line were connected to the microphone amplifier output. Breaking the transmission path also allows a station operator to monitor an intercom channel while talking on a radio channel.

3.16.1.2 Monitoring.-

3.16.1.2.1 Radio.- The equipment station shall provide capability for monitoring incoming signals from any of the three radios not selected for transmission. To monitor a radio channel the station operator shall operate the appropriate selector switch to the MONITOR position. The volume of an incoming monitor signal shall be attenuated 10 decibels (db.) relative to the normal incoming volume with the radio selector set to TALK position.

3.16.1.2.2 Intercom.- The equipment station shall provide the capability for monitoring an intercom channel while the station operator is communicating on a radio channel. To monitor an intercom channel the station operator shall set the INTERCOM selector switch to either the CHANNEL A or CHANNEL B position. (Outgoing speech from the station will not be transmitted on the selected intercom channel because the transmission path from the microphone amplifier to the intercom line is broken by operation of radio selector switch to TALK position.) With an intercom channel selected for monitoring and any one of the RADIO selectors in TALK position the level of the incoming intercom signal shall be attenuated 10 db below the normal incoming signal level for the operating conditions specified in 3.16.2. With an intercom channel selected for monitoring and all of the radio selector switches in either the OFF or MONITOR positions the intercom signal shall be received at the normal incoming signal level for operating conditions specified in 3.16.2.

3.16.2 Operation for communication on an intercom channel.- The equipment stations shall provide the capability for two-way communication on either of two intercom channels. Physically, each channel will consist of one twisted pair of wires available at all stations in the system. Operation of the equipment station for communication on an intercom channel shall be as specified in 3.16.2.1 through 3.16.2.3. (Operation is specified for CHANNEL A, operation for CHANNEL B shall be identical except INTERCOM selector switch is set to CHANNEL B.)

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3.16.2.1 The INTERCOM selector switch shall be set to CHANNEL A position. All other selector switches shall be set to OFF position.

3.16.2.2 The receiver amplifier input shall be connected through isolation resistors and a transformer to intercom CHANNEL A. Incoming signals from CHANNEL A shall be amplified and applied to the station headset.

3.16.2.3 Operation of the microphone press-to-talk switch shall cause the output of the microphone amplifier to be connected to intercom CHANNEL A. Microphone output signals shall be amplified and applied to intercom CHANNEL A.

3.17 Electrical requirements.-

3.17.1 Power supply.- The equipment shall be designed to operate as specified herein when connected to a 28 +2/-6 volt d.c. supply. The equipment shall be capable of operating for extended periods with any d.c. supply voltage in the 22 to 30 volt range. When the equipment is tested as specified in 4.6.6 the talk relay shall operate properly and the output signal harmonic distortion shall not exceed the limit specified in 3.18.1.6. Unless otherwise specified in the contract or order, the power supply voltage shall be 28 volts d.c. for the performance tests specified herein.

3.17.2 Circuit design.- The general configuration of the circuit shall be in accordance with the simplified schematic shown on figure 4. The detailed circuit design shall be in accordance with the schematic diagrams shown on figures 5 and 6 and shall be such that the equipment meets the performance requirements specified herein.

3.17.2.1 Input-output requirements for headset-microphone.- The circuit shall be designed to permit the use of a headset-microphone at each station. The headset earphones will be connected in parallel presenting a 10 ohm load to the receive amplifier output circuit. The microphone amplifier input circuit shall be designed to accept either a 5 ohm microphone or a 150 ohm microphone. Sensitivity of the amplifier for both types of microphone shall be as specified in 3.18.1.3. The wiring to the headset-microphone jack shall be such as to allow use of a headset-microphone with a 150 ohm microphone wired in accordance with figure 3 or a headset-microphone with a 5 ohm microphone wired in accordance with figure 2.

3.17.2.2 Input-output requirements for use with radio.- The equipment station shall accept audio input signals from any one of three radio receivers and amplify the signal to drive the station headset. Microphone signals originating from the station microphone shall be amplified by the equipment and applied to any one of three radio transmitters selected for transmission. Operating the microphone press-to-talk switch shall cause the transmit-receive control circuit of the selected radio to be completed which will switch the radio to the transmit mode of operation.

3.17.2.3 Input-output requirements for intercom operation.- The equipment station shall accept audio input signals from either of two intercom channels and amplify the signal to drive the station headset. Microphone signals originating from the station microphone shall be amplified by the equipment station and applied to either of the two intercom channels when selected for transmission.

3.17.3 Electrical performance requirements.-

3.17.3.1 Output signal to radio transmitter.- The equipment station shall be capable of producing an output voltage of 200 millivolts measured across the 600 ohm load when tested in accordance with 4.6.2.1.

3.17.3.2 Output signal to intercom line.- The equipment station shall be capable of producing an output voltage of 3 volts across the terminals for intercom CHANNEL A when tested in accordance with 4.6.2.2.

3.17.3.3 Output signal to headset.- The equipment station shall be capable of producing an output signal of 0.3 volts with less than 5 percent harmonic distortion across the 10 ohm resistance load when tested as specified in 4.6.3.1.1 or as specified in 4.6.3.1.2.

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3.17.3.3.1 Incoming volume control.- A front panel mounted volume control shall be provided for adjustment of headset volume for incoming radio, intercom, and sidetone signals. The control shall provide for continuous volume adjustment over a minimum range of 30 db.

3.17.3.4 Talk relay control circuit.- A talk relay shall be provided in the equipment station to control the microphone amplifier output, to control sidetone, and to provide switching of the radio from transmit to receive mode of operation. Figure 7 illustrates the switching necessary for control of radio transmitter. The talk relay shall be controlled by the microphone press-to-talk switch such that the relay is operated when the press-to-talk switch is closed. A pair of terminals shall be provided on the ship's wiring terminal board in the equipment which will allow use of an external foot switch in lieu of the microphone press-to-talk switch.

3.17.4 Interconnecting cables.- Type TTHFWA cable conforming to MIL-C-915 will be employed for interconnection of audio, control, and power circuits when installed. A maximum length of 1500 feet shall be used for calculating line impedance of audio and signal circuits. D.C. resistance of TTHFWA cable is 16 ohms per 1,000 feet.

3.18 Amplifiers.- The equipment shall contain two identical transistor amplifiers. One shall be used as a microphone amplifier and the other as a receive amplifier. The amplifiers shall be physically and electrically interchangeable, shall be plug-in types and may be encapsulated. Amplifier performance shall differ to the extent specified herein, depending upon application as either a microphone amplifier or a receive amplifier. Any circuit changes necessary to achieve the differences in performance shall be made external to the amplifier but within the equipment station.

3.18.1 Electrical characteristics.-

3.18.1.1 Input impedance.- Two microphone input circuits shall be provided. One input shall have an impedance of 150 ± 15 ohms for use with a 150 ohm microphone. The other input shall have a 5 ± 1 ohm impedance and shall be for use with a 5 ohm microphone.

3.18.1.2 Output impedance.- The amplifier output impedance shall be 100 ohms.

3.18.1.3 Input sensitivity.- The microphone amplifier shall be capable of producing rated output voltage as specified in 3.18.1.4. when driven, by either a 5 millivolt root mean square (r.m.s.) 1000 cycles per second (c.p.s.) signal applied to the 150 ohm input or a 200 microvolt r.m.s. 1000 c.p.s. signal applied to the 5 ohm input.

3.18.1.4 Rated output voltage.- The amplifier shall produce an output voltage of 3 volts r.m.s. across a 100 ohm noninductive resistance load when driven by either of the input signals specified in 3.18.1.3.

3.18.1.5 Frequency response.- The amplifier frequency response shall be within 3 db. of the response at 1000 c.p.s. for all frequencies in the range from 300 to 6000 c.p.s. when measured in accordance with 4.6.2.3.

3.18.1.6 Distortion.- The amplifier output signal shall contain not more than 5 percent harmonic distortion when the amplifier is producing rated output voltage under the test conditions specified in 4.6.2.4.

3.18.2 Amplifier noise and microphonics.- The amplifier noise and microphonics shall not exceed the following values when tested as specified in 4.6.2.5:

(a) Equipment stationary—10 millivolts
(b) Equipment vibrating (as required by MIL-I-983)—14 millivolts.

3.18.3 Stability.- The amplifiers shall show no evidence of instability or oscillation under any condition of operation.

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3.18.4 Endurance.- After being tested as specified in 4.6.4, the equipment shall be capable of normal operation in every respect. There shall be no visible damage to any electrical or mechanical component of the equipment.

3.19 Designation and marking.- Selector switch function and position designations shall be provided as shown on figure 1. The designations shall be illuminated as specified in 3.12.

3.20 Instruction plate and circuit diagram.- An instruction plate containing abridged operating instructions shall be provided on the outside of the unit. Schematic diagrams shall be provided and mounted inside the enclosure so as to be useable in servicing the unit. The circuit diagrams shall be of aluminum or plastic material.

3.21 Equipment repair parts.- Equipment repair parts shall be in accordance with MIL-E-17362.

3.22 Manuals.- The manual shall be type I in accordance with MIL-M-15071. The manual shall cover the equipment specified herein, and shall include headset-microphone operating techniques. The manual shall contain preliminary drawings, working drawings, and an illustrative schematic showing the interconnection of stations and a sufficient portion of each station to enable a reader to determine the operation of the switches and relays and to service and install the equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in table I, which shall be performed, in general, in the order listed.

Table I - Preproduction inspection

Inspection	Requirement paragraph	Test reference
General examination	3.3	MIL-I-983
Operating	3.16	4.6.1
Electrical performance	3.17.3	4.6.2 and 4.6.3
Supply line voltage variation	3.17.1	4.6.6
Shielding and radio frequency noise reduction	3.3	MIL-I-983
Amplifier noise and microphonics	3.18.2	4.6.2.5
Temperature and humidity	3.3	MIL-I-983
Insulation resistance	3.3	MIL-I-983
Endurance	3.18.4	4.6.4
Selector switch life test	3.10	4.6.5
Enclosure	3.3 and 3.5	MIL-I-983
Shocks, vibration and inclination	3.3	MIL-I-983

4.3 Sampling for quality conformance inspection.-

4.3.1 Lot.- All equipment presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.3.2 Sampling for group A tests.- Sample equipments shall be selected in accordance with MIL-STD-105 at inspection level III for the tests specified in 4.4. The acceptance quality level (AQL) shall be 1.5 percent.

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4.3.3 Sampling for group B tests.- Sample equipments shall be selected in accordance with table II for the tests specified in 4.4.

Table II - Sampling for group B tests.

Lot size number of equipment	Sample size number to be tested	Test failures	
		Number allowed	Reject the lot on
Up to 15	3	0	1
16 to 50	5	0	1
51 to 90	8	0	1
91 to 280	13	1	2
281 to 500	20	2	3
501 to 1200	32	3	4

4.3.4 Sampling for group C tests.- Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to the group C tests listed in 4.4.

4.4 Quality conformance inspection.- The sample equipments selected in accordance with 4.3 shall be subjected to the examination and tests specified in table III. Tests shall be performed, in general, in the order listed.

Table III - Quality conformance inspection

Inspection	Requirement paragraph	Test reference
Group A		
General examination	3.3	MIL-I-983
Operating	3.16	4.6.1
Electrical performance	3.17.3	4.6.2 and 4.6.3
Group B		
Supply line voltage variations	3.17.1	4.6.6
Amplifier noise and microphonics	3.18.2	4.6.2.5
Group C		
Shielding and radio frequency noise noise reduction	3.3	MIL-I-983
Endurance	3.18.4	4.6.4
Selector switch life test	3.10	4.6.5
Temperature and humidity	3.3	MIL-I-983
Insulation resistance	3.3	MIL-I-983
Enclosure	3.3 and 3.5	MIL-I-983
Shock, vibration and inclination	3.3	MIL-I-983

4.5 Rejection.- The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A, B or C test shall be counted as a defect, and the equipment shall not be offered for delivery.

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4.6 Test procedures.-

4.6.1 Operating test.-

4.6.1.1 Operation as radio control station.- The operation of the equipment station with a radio transmitter and receiver shall be tested for compliance with 3.16.1 using a simulated radio circuit.

4.6.1.2 Operation as an intercommunication station.- Two equipment stations shall be interconnected for normal intercommunication operation. Two-way communication shall be established on each of the two intercommunication channels. Operation shall be as specified in 3.16.2.

4.6.2 Microphone amplifier tests.-

4.6.2.1 Audio output voltage to transmitter.- The equipment shall be operated as specified in 3.16.1.1 with the RADIO 1 selector switch set to the TALK position and all other switches set to the OFF position. The output terminals for transmission to RADIO 1 shall be connected to a 600 ohm noninductive resistance load. The microphone press-to-talk relay control circuit shall be closed to energize the relay. A 200 microvolt r.m.s. 1000 c.p.s. signal shall be applied to the 5 ohm microphone input. The output signal shall be measured across the 600 ohm load and shall comply with 3.17.3.1.

4.6.2.2 Audio output voltage to intercom line.- The equipment shall be operated as specified in 3.16.2 with the INTERCOM selector switch set to CHANNEL A position and all radio selector switches set to the OFF position. The microphone press-to-talk relay control circuit shall be closed to energize the talk relay. A 200 microvolt r.m.s. 1000 c.p.s. signal shall be applied to the 5 ohm microphone input. The output signal shall be measured across the output terminals provided for connecting the station to intercom CHANNEL A. The microphone amplifier will be terminated in a 100 ohm resistance load located within the equipment station. The output voltage for intercom CHANNEL A shall be measured without additional loading of the output circuit. The output signal shall be in accordance with 3.17.3.2.

4.6.2.3 Frequency response.- The equipment station shall be operated as specified in 4.6.2.2. With the input signal amplitude remaining constant at 200 microvolts the input signal frequency shall be varied from 300 to 6000 c.p.s. Output voltage shall be measured to determine compliance with 3.18.1.5.

4.6.2.4 Harmonic distortion.- With the equipment operating under the test conditions specified in 4.6.2.3 the harmonic distortion produced by the amplifier shall be measured for signal frequencies of 400 and 1000 c.p.s. Results shall comply with 3.18.1.6.

4.6.2.5 Noise and microphonics.- The equipment shall be operated as specified in 4.6.2.2. The input signal source shall then be disconnected and a 5 ohm resistor connected across the 5 ohm input. Output voltage shall be measured to determine compliance with 3.18.2. The measurement shall be made under two conditions of test, once while the component is held fixed and again while the component is under vibration as required by MIL-I-983. For this test the equipment station shall be energized with a 28 volt d.c. supply having a 1 volt r.m.s. 180 c.p.s. a.c. ripple voltage superimposed on the d.c. supply.

4.6.3 Receive amplifier tests.-

4.6.3.1 Output voltage to headset.-

4.6.3.1.1 Input signal from radio receiver.- The equipment shall be operated as specified in 3.16.1.1 with the RADIO 1 selector switch set to the TALK position and all other switches set to the OFF position. The output terminals for the headset shall be loaded with a 10 ohm noninductive resistance load and the volume control shall be set to the maximum position. A 3 volt r.m.s. 1000 c.p.s. signal shall be applied to the station input terminals for the RADIO 1 receiver. The receiver amplifier output shall be measured across the 10 ohm load. Results shall comply with 3.17.3.3.

4.6.3.1.2 Input signal from intercom channel.- The equipment shall be operated as specified in 3.16.2 with the INTERCOM selector switch set to the CHANNEL A position and all other switches set to the OFF position. The output terminals for the headset shall be loaded with a 10 ohm noninductive resistance load and the volume control set to the maximum position the talk relay shall be de-energized. A 3

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volt r.m.s. 1000 c.p.s. signal shall be applied to the CHANNEL A intercom terminals of the station. The output signal shall be measured across the 10 ohm load. Results shall comply with 3.17.3.3. The volume control shall then be rotated to the minimum volume position to determine compliance with 3.17.3.3.1.

4.6.4 Endurance test.- The equipment shall be operated under the test conditions specified in 4.6.2.1 with the equipment energized by a 30 volt d.c. source. The relay control circuit shall be opened and closed to cause the relay to be energized for a period of 7 seconds and de-energized for a period of 15 seconds. This cycle shall be repeated for 20,000 continuous operations. Immediately upon completion of 20,000 operations the equipment shall be tested for normal operation. There shall be no noticeable change in the quality of speech transmitted, nor shall the output decrease more than 2 db. from the output measured at the beginning of the test.

4.6.5 Selector switch life test.- One of the radio selector switches of the station under test shall be used for this test. The switch shall be operated from the normal OFF position to the TALK position, then to the MONITOR position, and back to the OFF position. This cycle shall be repeated for 25,000 cycles. Upon completion of the test the equipment shall comply with 3.10.

4.6.6 Supply line voltage variation.- The equipment shall be operated as specified in 4.6.2.2. The power supply voltage shall be reduced from the normal 28 volts to a value of 22 volts. The equipment shall be checked for proper operation of the talk relay and the output signal harmonic distortion shall be measured for compliance with 3.18.1.6.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Intercommunication station.-

5.1.1.1 Preservation and packaging.- Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, manufacturer's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged by level A; packed by level C, and marked by levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing and marking.- The equipment and accessories, repair parts and technical publications shall be preserved and packaged by levels A or C; packed by levels A or B as specified, and marked in accordance with MIL-E-17555.)

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6. NOTES

6.1 Intended use. - The equipment covered by this specification is intended for intercommunication aboard Naval ships and communication over ship's radio circuits.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation and packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

6.3 Preproduction. - Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 5835-N103SH)

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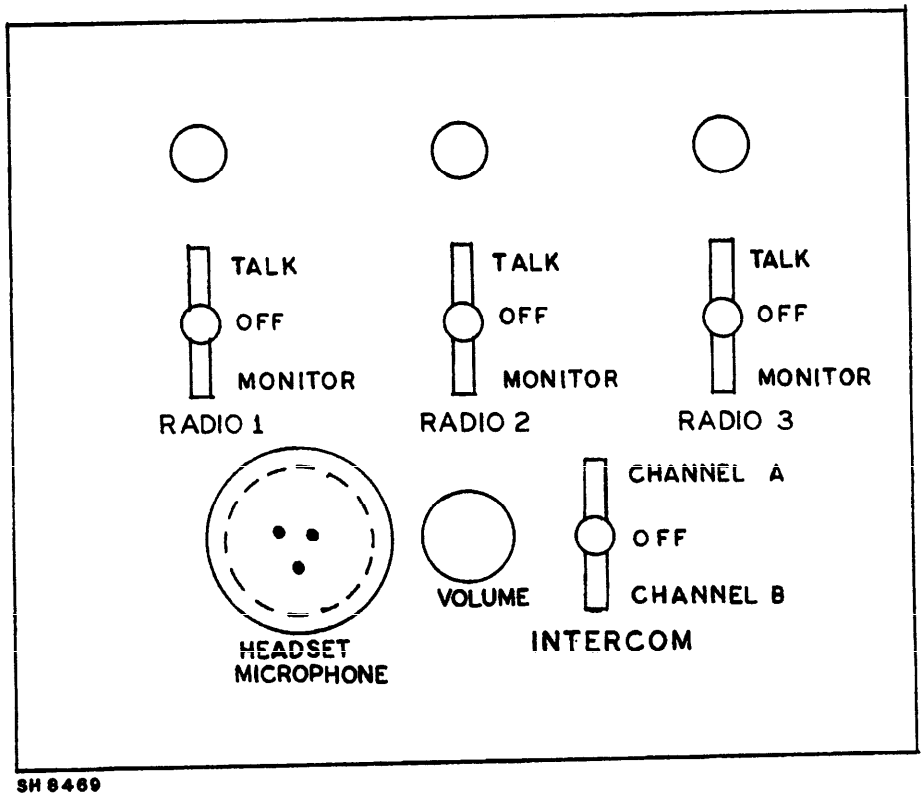


Figure 1 - Front panel

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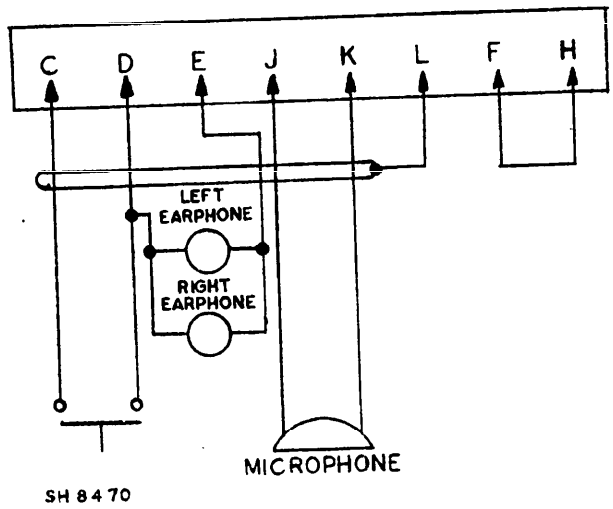


Figure 2 - Schematic for headset-microphone with 5-ohm microphone.

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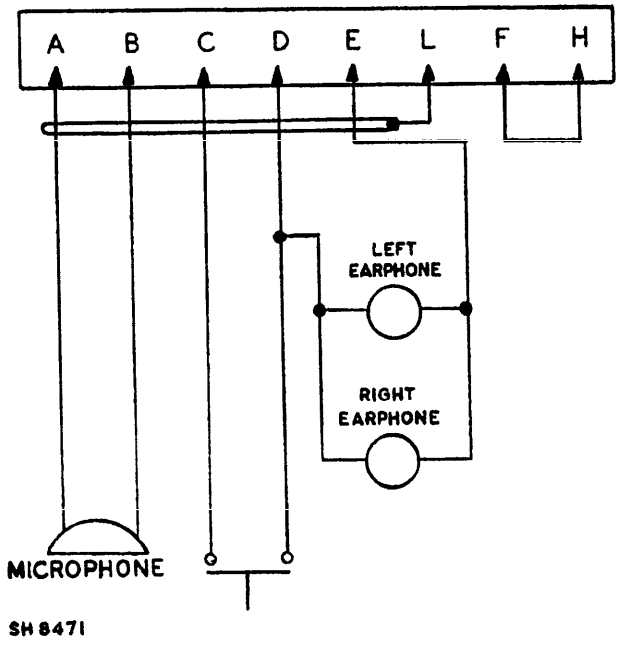


Figure 3 - Schematic for headset-microphone with 150-ohm microphone.

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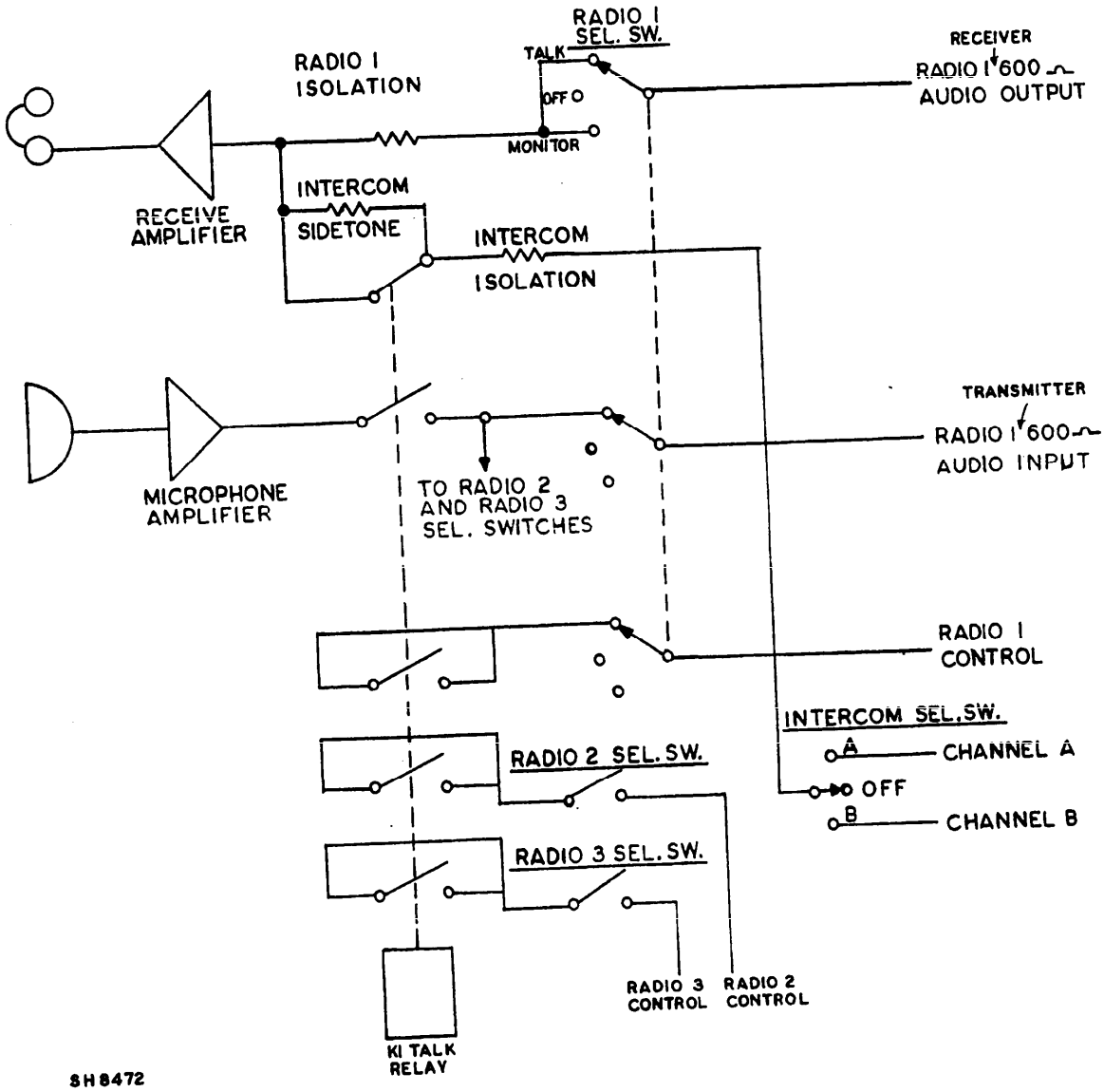


Figure 4 - Simplified schematic, equipment station set to transmit and receive on radio 1.

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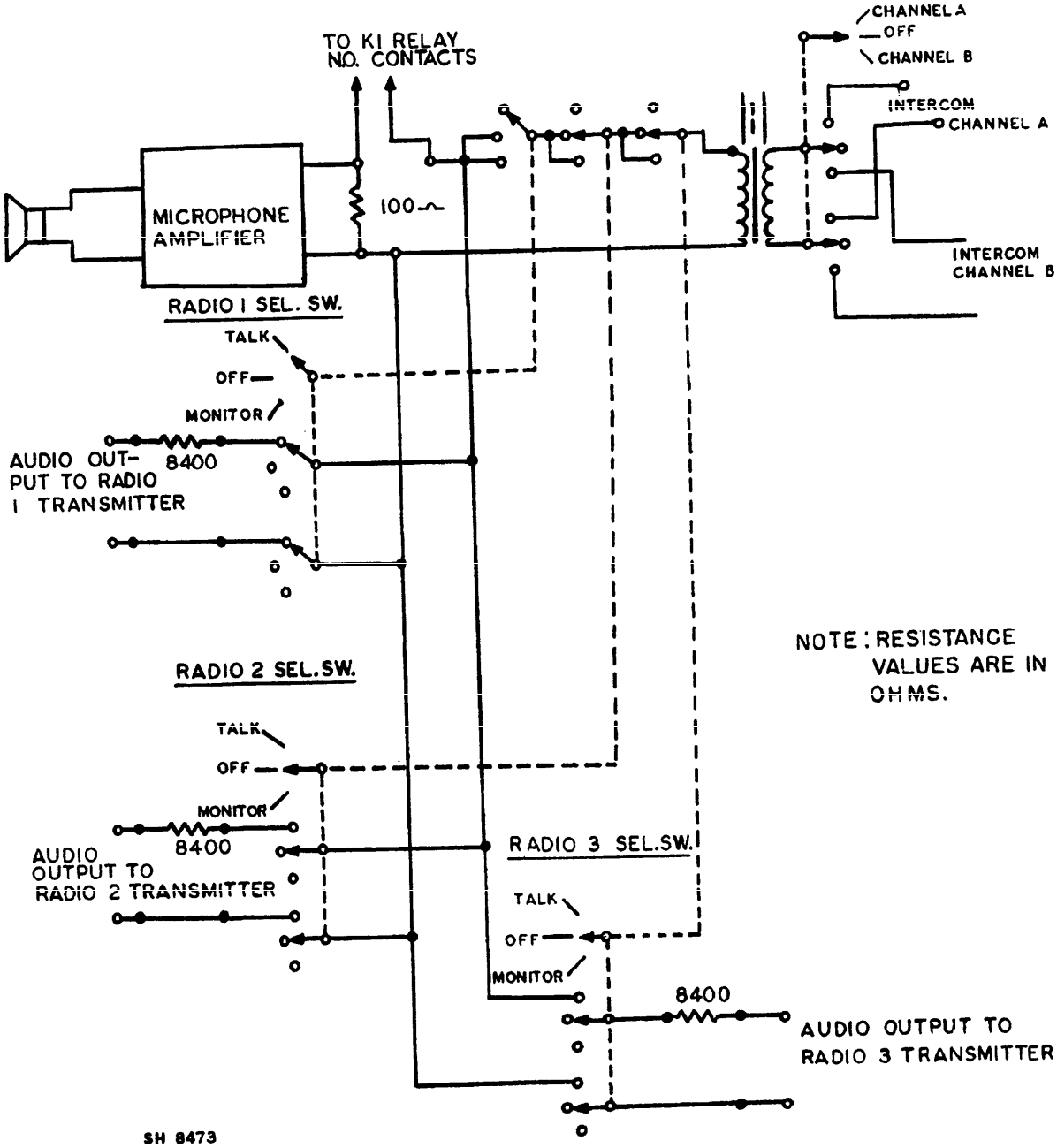


Figure 5 - Equipment station output circuit schematic.

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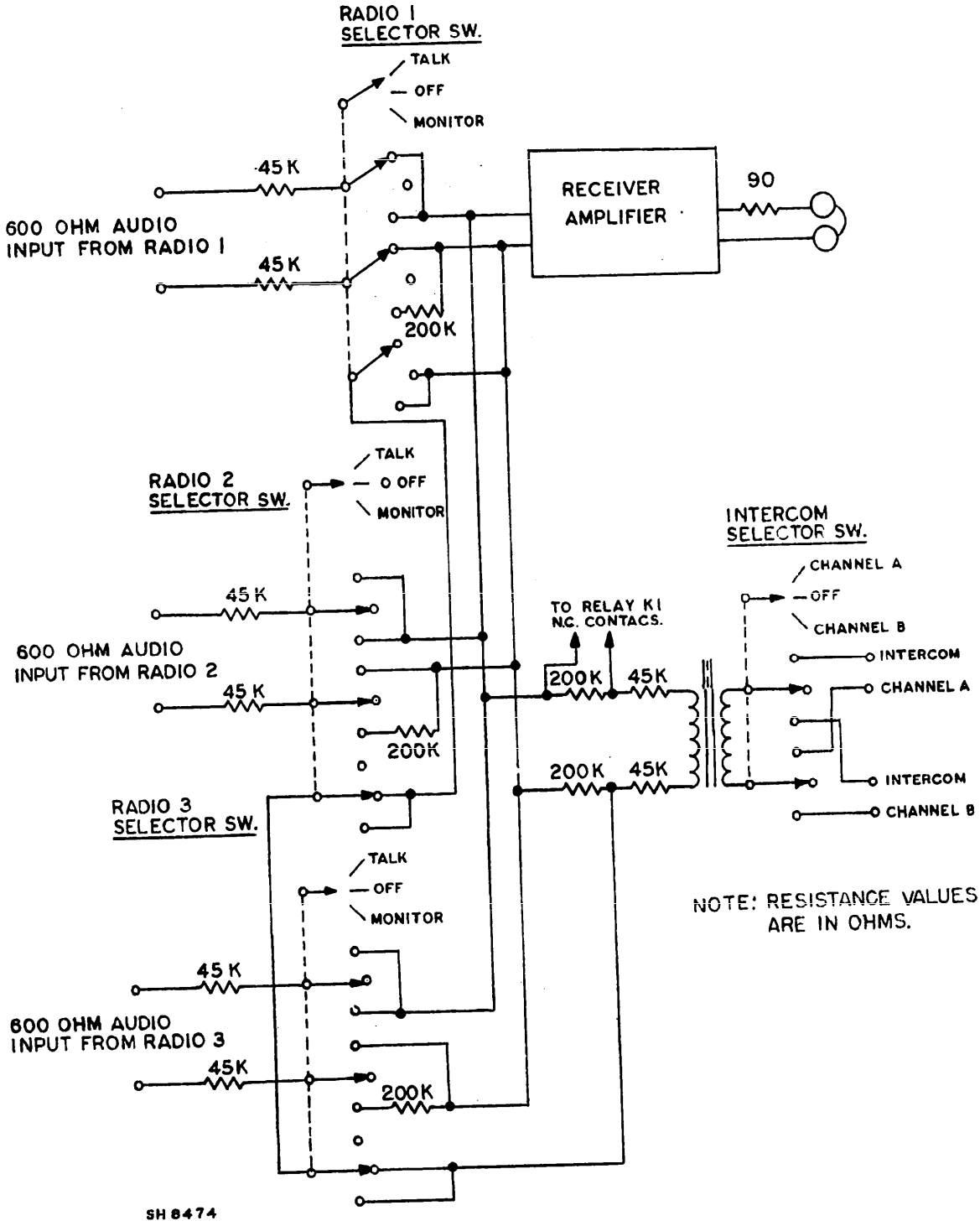


Figure 6 - Equipment station input circuit schematic.

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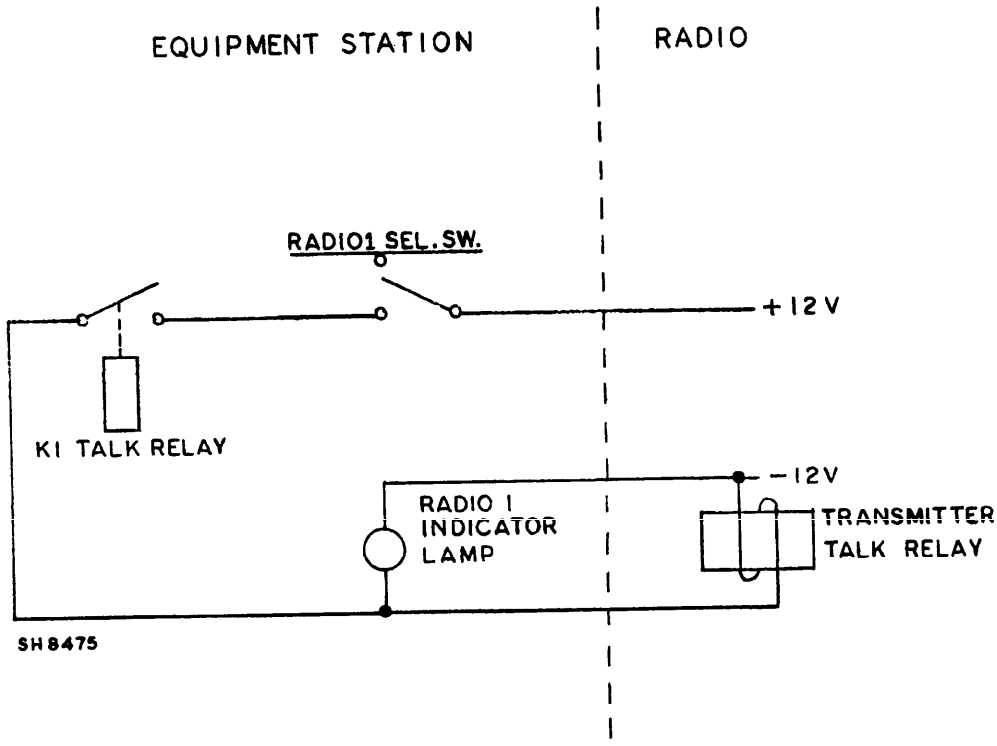


Figure 7 - Control circuit for radio transmitter.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE D14000

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